

Name: _____ (please print)

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ECE 2202 – Final Exam

April 30, 2025

**Keep this exam closed until you
are told to begin.**

1. This exam is closed book, closed notes. We will provide crib sheets.
2. Show all work on these pages. Show all work necessary to complete the problem. A solution without the appropriate work shown will receive no credit. A solution that is not given in a reasonable order will lose credit. Clearly indicate your answer (for example by enclosing it in a box).
3. Show all units in solutions, intermediate results, and figures. Units in the exam will be included between square brackets.
4. If the grader has difficulty following your work because it is messy or disorganized, you will lose credit.
5. Do not use red ink. Do not use red pencil.
6. You will have 120 minutes to work on this exam.

2. _____/20

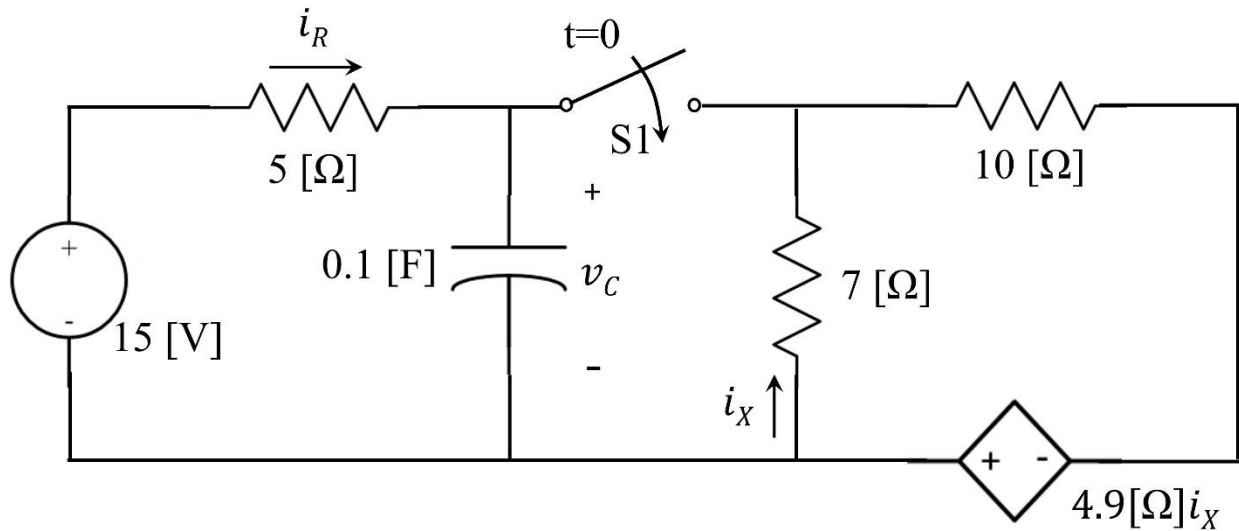
3. _____/30

4. _____/30

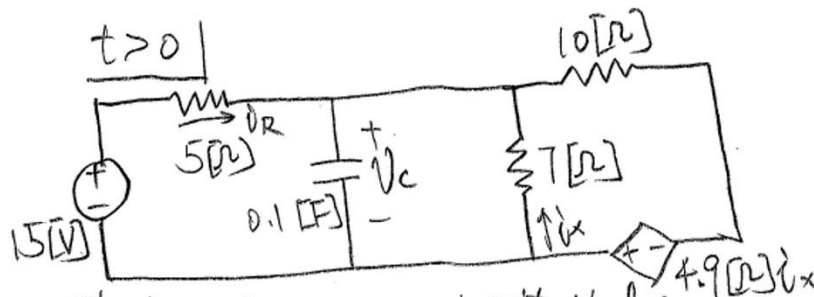
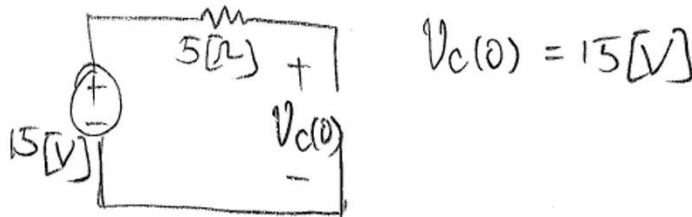
Total = 100

1. {20 Points} The switch S1 has been open for a long time before $t = 0$. The switch closes at $t = 0$. Please find:

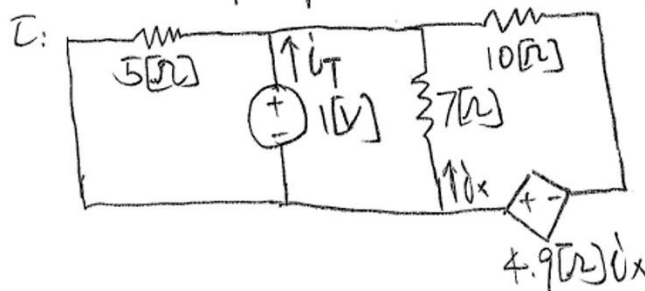
- 1) $v_C(t)$ for $t \geq 0$.
- 2) $i_R(0.25[s])$.



1) For $t < 0$



This is a step response circuit. Need to get R_{eq} seen by capacitor.



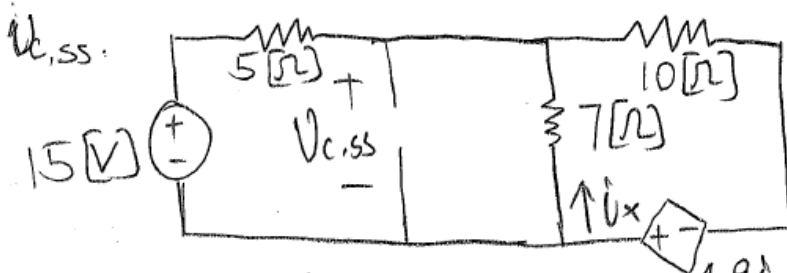
Use test source method,
set independent source
equal to "0".

Room for extra work

$$\begin{cases} \frac{1[V]}{5[\Omega]} - \dot{i}_T - \dot{i}_x + \frac{4.9\dot{i}_x + 1[V]}{10[\Omega]} = 0 \\ \dot{i}_x = -\frac{1[V]}{7[\Omega]} \end{cases}$$

Solve: $\begin{cases} \dot{i}_T = 0.3728 [A] \\ \dot{i}_x = -0.143 [A] \end{cases} \therefore R_{eq} = \frac{1[V]}{0.3728[A]} = 2.68 [\Omega]$

$$\therefore \tau = R_{eq} \cdot C = 2.68 \times 0.1 = 0.268 [s]$$



$$\begin{cases} \frac{v_{c,ss} - 15[V]}{5[\Omega]} - \dot{i}_x + \frac{v_{c,ss} + 4.9\dot{i}_x}{10[\Omega]} = 0 \\ \dot{i}_x = -\frac{v_{c,ss}}{7[\Omega]} \end{cases}$$

Solve: $\begin{cases} v_{c,ss} = 8.05 [V] \\ \dot{i}_x = -1.15 [A] \end{cases} \therefore \begin{aligned} v_c(t) &= 8.05 + (15 - 8.05)e^{-\frac{t}{0.268[s]}} \\ v_c(t) &= 8.05 + 6.95 e^{-\frac{t}{0.268[s]}} \quad (t \geq 0) \end{aligned}$

2) $v_c(0.25) = 8.05 + 6.95 \cdot e^{-\frac{0.25}{0.268}} = 10.78 [V]$

$$i_R(0.25[s]) = \frac{15[V] - v_c(0.25[s])}{5[\Omega]} = \frac{15 - 10.78}{5} = 0.844 [A]$$

2. {20 Points} Refer to Figure 1 below. The switch has been closed for a long time. At time $t=0$, the switch opens.

- Find $i_L(0^+)$.
- Find $v_C(0^+)$.
- Find $\frac{di_L(0^+)}{dt}$.
- Find $\frac{dv_C(0^+)}{dt}$.
- Please solve the RLC circuit in Figure 1 and find an expression for $v_C(t)$ for $t \geq 0$.

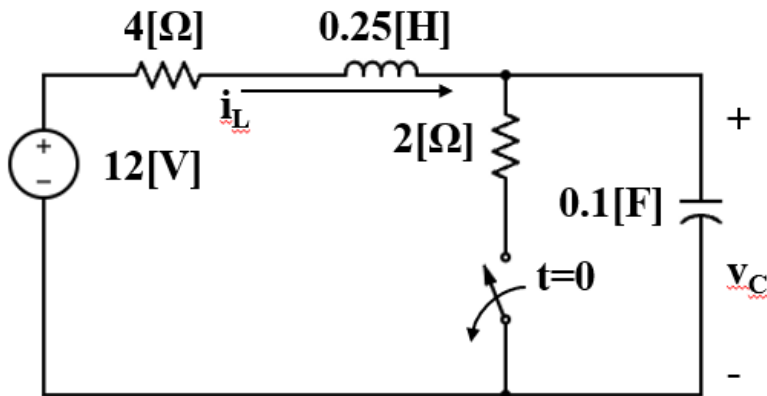


Figure 1
Figure 1

$t < 0$: 4Ω

$12V$ $\rightarrow i_L(0^-)$ 2Ω $v_C(0^-)$

$i_L(0^-) = 2A$

$v_C(0^-) = 12 \left(\frac{2}{6} \right) = 4V$

(a) $t \geq 0$: 4Ω $0.25H$ $0.1F$

$t = 0$: $12V$ $\rightarrow i_L(0^+)$ $v_C(0^+)$

$i_L(0^+) = i_L(0^-) = 2A$

(b) $v_C(0^+) = v_C(0^-) = 4V$

(c) $v_L(0^+) = L \frac{di_L(0^+)}{dt}$

$\frac{di_L(0^+)}{dt} = \frac{v_L(0^+)}{L}$

KVL:

$-12V + i_L(0^+)4\Omega + v_L(0^+) + v_C(0^+) = 0$

$-12V + 8V + v_L(0^+) + 4V = 0$

$v_L(0^+) = 0V$

$\frac{di_L(0^+)}{dt} = 0 \left[\frac{A}{s} \right]$

Room for extra work

$$(d) \frac{dv_c(0^+)}{dt} = \frac{i_c(0^+)}{C}$$

$$i_c(0^+) = i_L(0^+) = 2 \text{ [A]}$$

$$\frac{dv_c(0^+)}{dt} = \frac{2 \text{ [A]}}{0.1 \text{ [F]}} = 20 \left[\frac{\text{V}}{\text{s}} \right]$$

$$(e) \alpha = \frac{R}{2L} = \frac{4}{0.5} = 8 \left[\frac{\text{rad}}{\text{s}} \right]$$

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{(0.25)(0.1)}} = 6.3246 \left[\frac{\text{rad}}{\text{s}} \right]$$

$\alpha > \omega_0$ overdamped.

$$s_{1,2} = -8 \pm \sqrt{8^2 - (6.3246)^2}$$

$$= -3.1010, -12.8989$$

$$A_1 = \frac{20 - (-12.8989)(4-12)}{-3.1010 + 12.8989}$$

$$A_2 = \frac{20 - (-3.1010)(4-12)}{-12.8989 + 3.1010}$$

$$A_1 = -8.4907$$

$$A_2 = 0.4907$$

$$v_c(t) = -8.4907 e^{-3.101t} + 0.4907 e^{-12.8989t} + 12 \text{ [V]}$$

$$A_1 = \frac{i_c(0^+) - s_2(v_c(0^+) - v_c(\infty))}{s_1 - s_2}$$

$$A_2 = \frac{i_c(0^+) - s_1(v_c(0^+) - v_c(\infty))}{s_2 - s_1}$$

$$v_c(\infty) = 12 \text{ [V]}$$

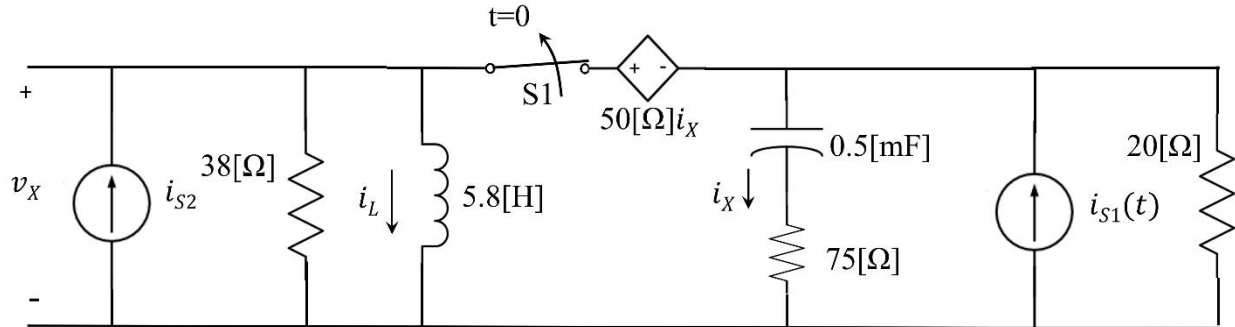
3. {30 Points} The circuit shown below has been in steady-state for a long time before $t = 0$, at $t = 0$, switch S1 opens.

$$i_{S1}(t) = 3.3 \cdot \sin\left(50 \left[\frac{\text{rad}}{\text{s}}\right] t + 150^\circ\right) [\text{A}]$$

$$i_{S2} = 0.05 [\text{A}]$$

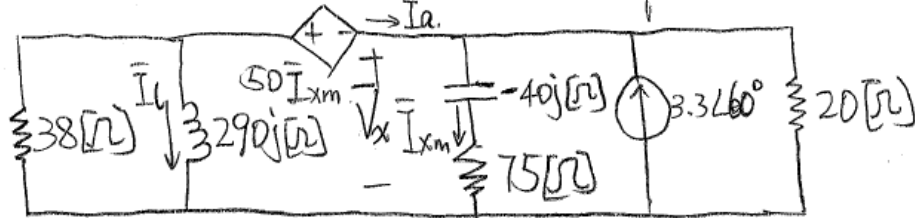
Please:

- 1) For $t \leq 0$, please use phasor analysis to solve the circuit and find $i_L(0)$.
- 2) Find $v_X(t)$ for $t > 0$.



1) Use superposition principle.

Take i_{S1} , set $i_{S2} = 0$, convert the circuit to phasor domain:



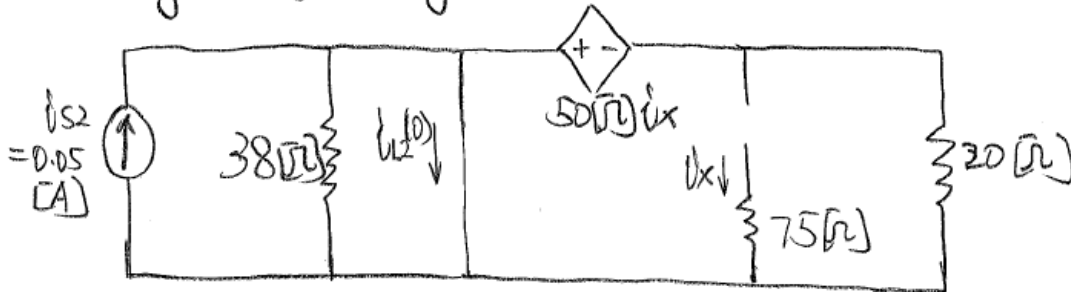
$$\begin{cases} -\bar{I}_a + \bar{I}_{xm} - 3.3 \angle 60^\circ + \frac{\bar{V}_x}{20} = 0 \\ \frac{\bar{V}_x}{75 - 40j} = \bar{I}_{xm} \\ \frac{\bar{V}_x + 50 \cdot \bar{I}_{xm}}{290j} + \frac{\bar{V}_x + 50 \bar{I}_{xm}}{38} + \bar{I}_a = 0 \\ \bar{I}_L = \frac{\bar{V}_x + 50 \bar{I}_{xm}}{290j} \end{cases} \quad \text{solve:} \quad \begin{cases} \bar{I}_a = -0.694 - 1.136j \\ \bar{I}_{xm} = 0.0413 + 0.38j \\ \bar{V}_x = 18.295 + 26.84j \\ \bar{I}_L = 0.158 - 0.07j \\ = 0.172 \angle -23.8^\circ [\text{A}] \end{cases}$$

$$\therefore i_L(t) = 0.172 \cdot \cos\left(50 \left[\frac{\text{rad}}{\text{s}}\right] t - 23.8^\circ\right) [\text{A}]$$

$$i_L(0) = 0.172 \cdot \cos(-23.8^\circ) = 0.157 [\text{A}]$$

Room for extra work

Take i_{s2} , set $i_{s1} = 0$: i_{s2} is a D.C. source, circuit has been in steady-state for a long time:

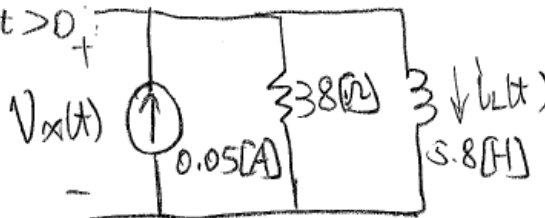


$i_x = 0$. $\therefore 50 i_x = 0$: dependent source is a wire.

$$i_{L2}(0) = 0.05 \text{ [A]}$$

$$\therefore i_L(0) = i_{L1}(0) + i_{L2}(0) = 0.157 + 0.05 = 0.207 \text{ [A]}$$

2) For $t > 0$:



step response circuit:

$$\tau = \frac{L}{R} = \frac{5.8 \text{ [H]}}{38 \text{ [}\Omega\text{]}} = 0.152 \text{ [s]}$$

$$i_{L,ss} = 0.05 \text{ [A]}$$

$$i_L(0) = 0.207 \text{ [A]}$$

$$i_L(t) = 0.05 + (0.207 - 0.05) e^{-\frac{t}{0.152}} \quad (t \geq 0)$$

$$= 0.05 + 0.157 e^{-\frac{t}{0.152 \text{ [s]}}} \quad (t \geq 0)$$

$$\therefore V_x(t) = L \frac{di_L(t)}{dt} = 5.8 \times 0.157 \left(-\frac{1}{0.152}\right) e^{-\frac{t}{0.152}}$$

$$V_x(t) = -5.99 e^{-\frac{t}{0.152 \text{ [s]}}} \quad (t > 0)$$

4. {30 Points} Refer to Figure 2 below. All circuits are operating in steady-state.
- For Figure 2A, find the equivalent impedance with respect to terminals a and b.
 - For Figure 2A, find the open circuit voltage with respect to terminals a and b.
 - Using your solutions from part a and part b, draw and label the Thevenin equivalent circuit of Figure 2A with respect to terminals a and b by filling in the missing info of Figure 2B.
 - Next, the circuit shown in Figure 2B is connected to three parallel loads as shown in Figure 2C. After this connection is made, the following information is obtained:
 The rms voltage $|\overline{V}_{L,rms}|$ measured from node c to node d is 500 [V_{rms}].
 Load $\overline{Z}_1$ absorbs 10 [kW] at a lagging power factor of 0.8.
 Load $\overline{Z}_2$ absorbs 20 [kW] and -15 [kVAR].
 Load $\overline{Z}_3$ absorbs 30 [kVA] at a leading power factor of 0.5.
 - What is the complex power delivered by the voltage source $\overline{V}_{Th}$?

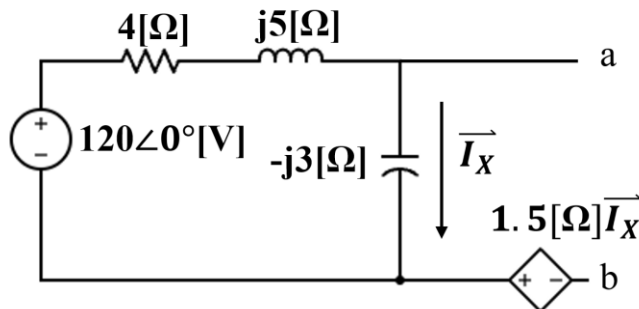


Figure 2A

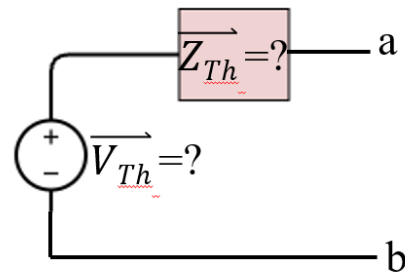


Figure 2B

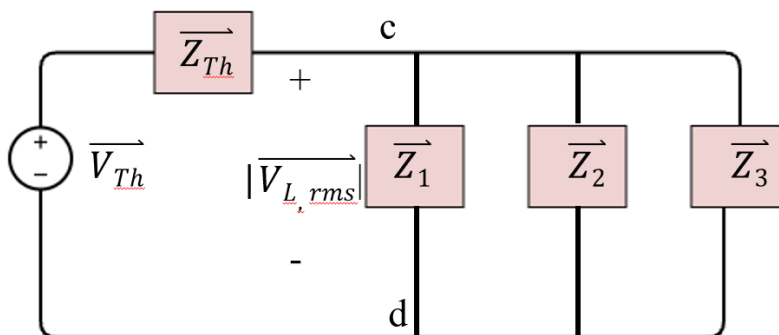
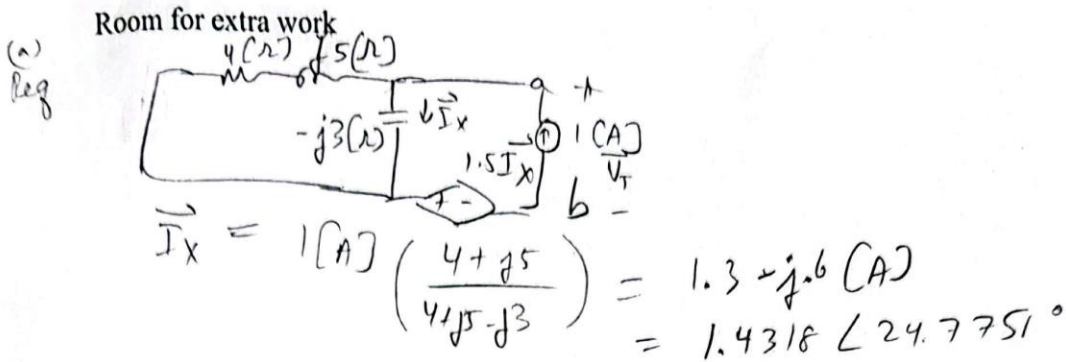


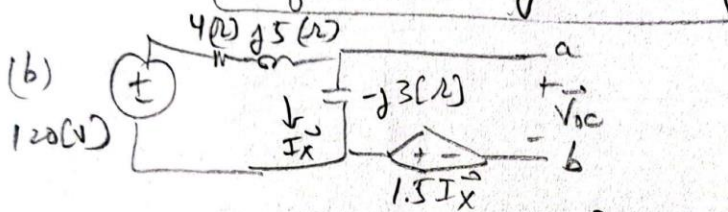
Figure 2C



$$\vec{V}_T - \vec{I}_x(-j3) - 1.5\vec{I}_x = 0$$

$$\vec{V}_T = 3.75 - j3 [V] = 4.8023 \angle -38.6598^\circ [V]$$

$$\vec{Z}_{eq} = 3.75 - j3 [\Omega]$$

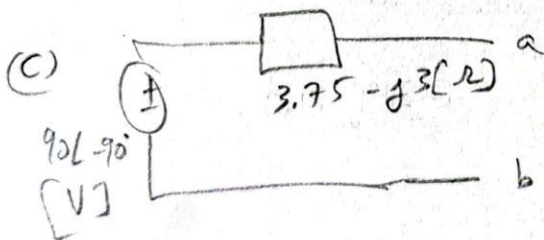


$$-120[V] + (4 + j5)\vec{I}_x + (-j3)\vec{I}_x = 0$$

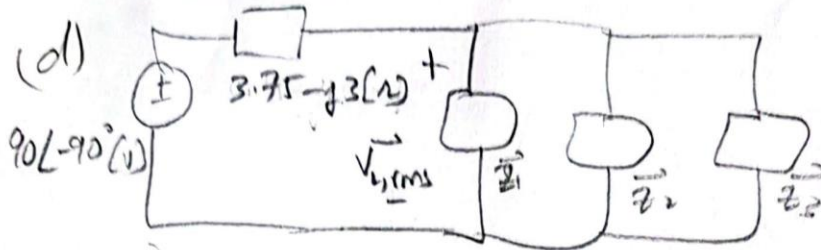
$$\vec{I}_x = 24 - j12 [A] = 26.8328 \angle -26.5651^\circ [A]$$

$$\vec{V}_{oc} - 1.5\vec{I}_x - \vec{I}_x(-j3) = 0$$

$$\vec{V}_{oc} = -j90 [V] = 90 \angle -90^\circ [V]$$



Room for extra work



$$P_1 = 10000 \text{ [W]}$$

$$\theta_1 = \cos^{-1}(0.8) = 36.8699^\circ$$

$$|\vec{S}_1| = \frac{P_1}{\cos\theta} = 12500 \text{ [VA]}$$

$$Q_1 = |\vec{S}_1| \sin\theta = 7500 \text{ [VAR]}$$

$$\vec{S}_1 = 10000 + j7500 \text{ [VA]}$$

$$\vec{S}_2 = 20000 - j15000 \text{ [VA]}$$

$$\theta_3 = \cos^{-1}(0.5) = 60^\circ \rightarrow \text{leading}$$

$$\vec{S}_3 = 15000 + j30000 \sin(-60^\circ)$$

$$\vec{S}_3 = 15000 - j25981 \text{ [VA]}$$

$$\vec{S}_{\text{tot}} = \vec{S}_1 + \vec{S}_2 + \vec{S}_3 = 45000 - j33481 \text{ [VA]}$$

$$= 56089 \angle -36.6499^\circ \text{ [VA]}$$

$$= |\vec{I}_{\text{rms}}| |\vec{V}_{\text{rms}}| \angle \phi_r - \phi_i$$

$$= |\vec{I}_{\text{rms}}| 500 \text{ [V]} \angle \phi_r - \phi_i$$

$$56089 = |\vec{I}_{\text{rms}}| 500 \text{ [V]}$$

$$|\vec{I}_{\text{rms}}| = 112.1777 \text{ [A]}$$

$$\vec{S}_{\text{Zeq}} = |\vec{I}_{\text{rms}}|^2 \vec{Z}_{\text{eq}} = 47189 - j77752 \text{ [VA]}$$

$$\vec{S}_{\text{Vth}} = \vec{S}_{\text{tot}} + \vec{S}_{\text{Zeq}} = 92189 - j71232 \text{ [VA]}$$

$$= 111650 \angle -37.6923^\circ \text{ [VA]}$$

$$v = iR \quad R_1 || R_2 = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2}} = \frac{R_1 R_2}{R_1 + R_2} \quad p = Cv \frac{dv}{dt} \quad p = Li \frac{di}{dt}$$

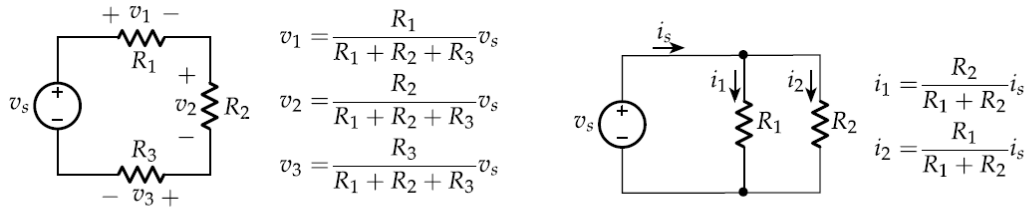
$$p = iv = i^2 R = \frac{v^2}{R} \quad w = \frac{1}{2} Li^2 \quad w = \frac{1}{2} Cv^2$$

$$R = \frac{v}{i}$$

$$v_C(t) = v_C(\infty) + [v_C(T_0) - v_C(\infty)] e^{-(t-T_0)/\tau}$$

$$i_L(t) = i_L(\infty) + [i_L(T_0) - i_L(\infty)] e^{-(t-T_0)/\tau}$$

(Switch action at $t = T_0$)



$$v_B = \underbrace{V_B}_{\text{DC}} + \underbrace{v_b}_{\text{AC}}$$

$\bar{V}_b \rightarrow$ Phasor notation

$$CMRR = 20 \log_{10} \frac{|A_d|}{|A_{cm}|}$$

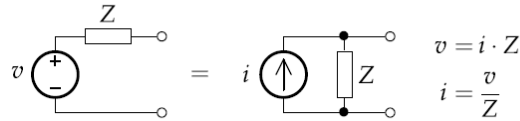
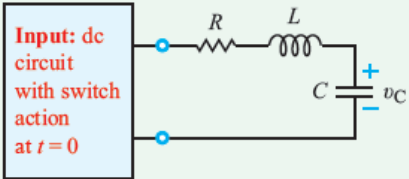
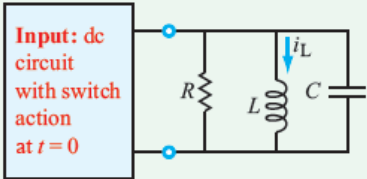


Table 6-2: General solution for second-order circuits for $t \geq 0$.

$x(t)$ = unknown variable (voltage or current)	
Differential equation:	$x'' + ax' + bx = c$
Initial conditions:	$x(0)$ and $x'(0)$
Final condition:	$x(\infty) = \frac{c}{b}$
$\alpha = \frac{a}{2}$	$\omega_0 = \sqrt{b}$
Overdamped Response $\alpha > \omega_0$	
$x(t) = [A_1 e^{s_1 t} + A_2 e^{s_2 t} + x(\infty)] u(t)$	
$s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2}$	$s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2}$
$A_1 = \frac{x'(0) - s_2[x(0) - x(\infty)]}{s_1 - s_2}$	$A_2 = -\left[\frac{x'(0) - s_1[x(0) - x(\infty)]}{s_1 - s_2} \right]$
Critically Damped $\alpha = \omega_0$	
$x(t) = [(B_1 + B_2 t) e^{-\alpha t} + x(\infty)] u(t)$	
$B_1 = x(0) - x(\infty)$	$B_2 = x'(0) + \alpha[x(0) - x(\infty)]$
Underdamped $\alpha < \omega_0$	
$x(t) = [(D_1 \cos \omega_d t + D_2 \sin \omega_d t) e^{-\alpha t} + x(\infty)] u(t)$	
$D_1 = x(0) - x(\infty)$	$D_2 = \frac{x'(0) + \alpha[x(0) - x(\infty)]}{\omega_d}$
$\omega_d = \sqrt{\omega_0^2 - \alpha^2}$	

Table 6-1: Step response of RLC circuits for $t \geq 0$.

Series RLC	Parallel RLC
Input: dc circuit with switch action at $t = 0$ 	Input: dc circuit with switch action at $t = 0$ 
Total Response	Total Response
Overdamped ($\alpha > \omega_0$) $v_C(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t} + v_C(\infty)$ $A_1 = \frac{\frac{1}{C} i_C(0) - s_2 [v_C(0) - v_C(\infty)]}{s_1 - s_2}$ $A_2 = \frac{\frac{1}{C} i_C(0) - s_1 [v_C(0) - v_C(\infty)]}{s_2 - s_1}$	Overdamped ($\alpha > \omega_0$) $i_L(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t} + i_L(\infty)$ $A_1 = \frac{\frac{1}{L} v_L(0) - s_2 [i_L(0) - i_L(\infty)]}{s_1 - s_2}$ $A_2 = \frac{\frac{1}{L} v_L(0) - s_1 [i_L(0) - i_L(\infty)]}{s_2 - s_1}$
Critically Damped ($\alpha = \omega_0$) $v_C(t) = (B_1 + B_2 t) e^{-\alpha t} + v_C(\infty)$ $B_1 = v_C(0) - v_C(\infty)$ $B_2 = \frac{1}{C} i_C(0) + \alpha [v_C(0) - v_C(\infty)]$	Critically Damped ($\alpha = \omega_0$) $i_L(t) = (B_1 + B_2 t) e^{-\alpha t} + i_L(\infty)$ $B_1 = i_L(0) - i_L(\infty)$ $B_2 = \frac{1}{L} v_L(0) + \alpha [i_L(0) - i_L(\infty)]$
Underdamped ($\alpha < \omega_0$) $v_C(t) = e^{-\alpha t} (D_1 \cos \omega_d t + D_2 \sin \omega_d t) + v_C(\infty)$ $D_1 = v_C(0) - v_C(\infty)$ $D_2 = \frac{\frac{1}{C} i_C(0) + \alpha [v_C(0) - v_C(\infty)]}{\omega_d}$	Underdamped ($\alpha < \omega_0$) $i_L(t) = e^{-\alpha t} (D_1 \cos \omega_d t + D_2 \sin \omega_d t) + i_L(\infty)$ $D_1 = i_L(0) - i_L(\infty)$ $D_2 = \frac{\frac{1}{L} v_L(0) + \alpha [i_L(0) - i_L(\infty)]}{\omega_d}$
Auxiliary Relations $\alpha = \begin{cases} \frac{R}{2L} & \text{Series RLC} \\ \frac{1}{2RC} & \text{Parallel RLC} \end{cases}$ $s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2}$	
$\omega_0 = \frac{1}{\sqrt{LC}}$ $\omega_d = \sqrt{\omega_0^2 - \alpha^2}$ $s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2}$	

$$v(t) = V_m \cos(\omega t + \theta_v); i(t) = I_m \cos(\omega t + \theta_i);$$

$$p(t) = \frac{V_m I_m}{2} \cos \theta + \frac{V_m I_m}{2} \cos \theta \cos(2\omega t) - \frac{V_m I_m}{2} \sin \theta \sin(2\omega t);$$

$$P = p_{ave} = \frac{V_m I_m}{2} \cos \theta; Q = \frac{V_m I_m}{2} \sin \theta; \theta = \theta_v - \theta_i; Z = \frac{V}{I} = \frac{V_m}{I_m} \angle \theta;$$

$\cos \theta$: power factor, p.f.; $\sin \theta$: reactive factor, r.f.;

θ : power factor angle;

$S = P + jQ$; Unites: S: [VA]; P: [W]; Q: [VAR];

$$S = \frac{V_m I_m^*}{2} = V_{rms} I_{rms}^* ;$$

$$P = |I_{rms}|^2 R; Q = |I_{rms}|^2 X; Z = R + jX;$$

$$S = \frac{|V_{rms}|^2}{R - jX};$$

